Biphasic Electrostimulation Artifact Model for ECAP Extraction in Spinal Cord Stimulation

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***Abstract*— Evoked compound action potential (ECAP) is extensively used as biomarker for neural activity in closed-loop spinal cord stimulation (SCS). However, accurate monitoring of neural activity is challenged by the low signal-to-noise ratio of ECAP recording, primarily due to the interference from stimulation-induced artifacts. This study proposes biphasic electrostimulation artifact model (BEAM) for ECAP extraction and assesses its efficacy using an enhanced quantification approach. The BEAM model is built based on the principle of additivity and incorporates boundary conditions that account for the equivalent electrical environment and temporal relationships. A bilinear growth curve model is introduced to calculate the E-score, enabling a comparison of ECAP extraction results with other methods. The proposed approach is extensively validated using clinical data from nine-month SCS therapy. BEAM method achieves the highest E-scores and reduces stimulation artifacts clinically without distorting information related to neural activity. BEAM accurately characterizes the morphology of artifacts generated by stimulation across various neurostimulation scenarios. Our model may lead to unique interpretation of neural activation in ECAP signals, evoked by electrical stimulation especially in closed-loop controlling strategies that aim to maintain consistent neural excitation.**

***Index Terms*— Artifact, ECAP extraction, evoked compound action potential, spinal cord stimulation.**

I. Introduction

SPINAL cord stimulation (SCS) is one of the important management for chronic pain, with approximately 34,000 implants performed worldwide every year [1], [2]. SCS alters neural activity through the targeted delivery of a stimulus to the dorsal column of the spinal cord [3]. However, its effects vary depending on the position of the leads relative to the neural targets [2]. Body movements and posture changes shift the position of the spinal cord within the spinal canal [4]. This can lead to changes in the recruitment of neural fibers, potentially causing discomfort or reducing the pain relief effect for the patient.

The closed-loop paradigm has been explored as a solution to this limitation [5], [6]. It adjusts the stimulation intensity based on monitoring stimulation-related biomarkers, ensuring the consistency of therapeutic effects [7]. Evoked Compound Action Potential (ECAP) is a widely studied biomarker [8], [9], [10], as illustrated in Figure 1 A, representing the aggregated response of regional nerve fibers to electrical stimulation [11]. Its amplitude, typically measured as the peak-to-peak value, is commonly used to quantify the effectiveness of stimulation.

However, it is inevitable that the measured ECAP signal is contaminated by stimulation artifact. The stimulation artifact refers to the residual decaying component that remains after the stimulation pulse, as illustrated in Figures 1B and E. Such artifacts lead to inaccurate estimation of the ECAP amplitude [12]. Notably, the strength of artifacts is usually orders of magnitude greater than that of the ECAP, making it critical to distinguish the ECAP from the concurrent stimulation artifact [13].To remove artifact, various extraction methods have been proposed, including artifact modeling [14], high-pass filtering [10] and template matching [15]. However, these methods effectiveness in SCS scenarios has not been thoroughly compared [10].

This work aims to reduce stimulation artifact in measured ECAP signals obtained under biphasic stimulation. Biphasic stimulation is clinically preferred due to its safety advantages [6], [16]. As illustrated in Figure 1B to D, it delivers consecutive positive and negative pulses thereby reducing the risk of damage to biological tissue and electrode interface by preventing charge accumulation [17]. However, the overlapping artifacts from two pulses complicate the morphology of measured signal. Obtaining accurate neural recording against artifacts still remains a challenge, and there is no consensus on ECAP processing method in biphasic stimulation [18].

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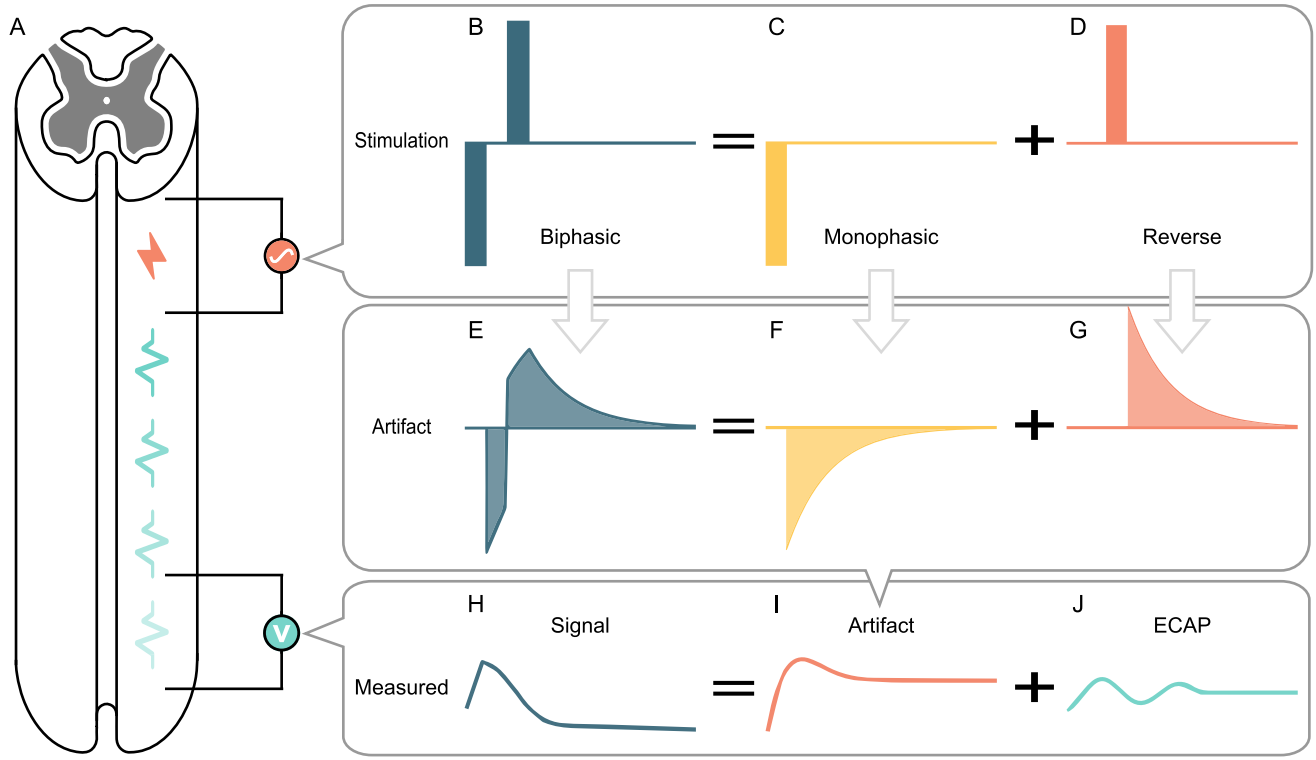


Fig. 1. Schematic illustrating the additivity of stimulation artifact in evoked compound action potential (ECAP) measurement during epidural spinal cord stimulation. (A) Neural activity of dorsal spinal cord (in blue) is evoked by epidural electrical stimulation (in orange) and measured by downstream electrodes. (B, C and D) Stimulation of biphasic waveform (B) can be considered as the addition of a monophasic pulse (C) and its reverse one (D). (E, F, G) The artifacts results from biphasic stimulation (E) can be deconstructed as the combination of the negative (F) and positive artifact (G), which are caused by monophasic pulse stimulation (C) and its reverse one (D) respectively. (H, I and J) Measured signal (H) consists of stimulation artifact (I) and ECAP (J). Note that these traces are illustrative only and are not shown to scale.

To extract ECAP under biphasic stimulation, we propose the biphasic electrostimulation artifact model (BEAM). Firstly, we develop BEAM with physically-based strategy, and optimize the evaluating method of growth curve analysis with E-score metric in Section II. In section III we describe the clinical experiment setup of SCS volunteers. Finally, section IV showcases the ECAP processing results achieved with BEAM, and compares the extracted ECAP and their evaluations with other methods. Furthermore, the generalizability of BEAM is validated in a DBS scenario, where stimulation artifacts interfere with neural recordings [19].

II. Artifact Model and Evaluation Method

In this section, we model the stimulation artifacts based on additive properties. Then we propose a modified denoising evaluation strategy grounded in signal processing interpretation.

A. Biphasic Electrostimulation Artifact Model

The artifact model is built to characterize artifact's shape. As illustrated in Figure 1H to J, the signal measured after epidural spinal cord stimulation consists of residual artifact and ECAP waveform. Here, we develop BEAM inspired by the strategy of alternating stimulation. Similarly, the electric field of biphasic stimulation in biological tissue is approximate to a linear, additive system [20], [21]. Consequently, the stimulation artifact of biphasic stimulation is assumed as the linear superposition of artifacts from two monophasic pulses with different onset time, as shown in Figure 1E to G. Then BEAM can be modified from classical double exponential model considering several physical boundary conditions.

1) The Classical Double Exponential Model: To express the artifacts in ECAP extraction during biphasic stimulus, as illustrated in Figure 2A, a typical double exponential model $f_0(t)$ is formulated as [15]

$$f_0(t) = \left(a_1 \cdot e^{-\frac{t}{b_1}} + c_1\right) + \left(a_2 \cdot e^{-\frac{t}{b_2}} + c_2\right), \tag{1}$$

where $t \in \mathbb{R}_{\geq 0}$ is time, parameters $a_{\{1,2\}}$, $b_{\{1,2\}}$, $c_{\{1,2\}}$ are factors that control the amplitude of the exponential, the decaying rate and the bias, respectively.

Because the artifact induced by biphasic stimulation consists of two monophasic stimulation, as shown in Figure 1B to G, the general model of biphasic stimulation artifact $f_3(t)$ is expressed as

$$f_3(t) = f_{\text{Mono}_1}(t) + f_{\text{Mono}_2}(t), \tag{2}$$

where function $f_{\text{Mono}_{\{1,2\}}}(t)$ represents monophasic artifact defined as $f_{\text{Mono}_{\{1,2\}}}(t) := a_{\{1,2\}} \cdot \exp(-t/b_{\{1,2\}}) + c_{\{1,2\}}$. Subscripts 1 and 2 indicate two individual artifacts, respectively. The mathematical formalization of $f_{\text{Mono}_{\{1,2\}}}(t)$ is given in Appendix-A.

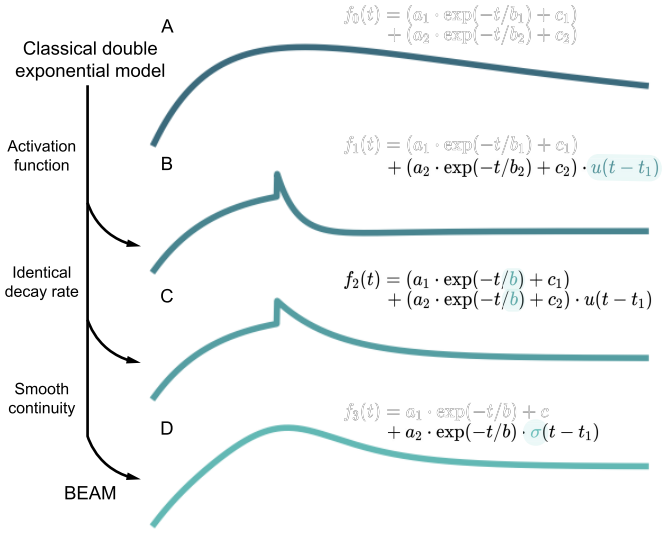


Fig. 2. Biphasic electrostimulation artifact model (BEAM) $f_3(t)$ modified from classical double exponential model $f_0(t)$ with boundary constraints. (A) The classical double exponential model $f_0(t)$ is composed of two arbitrary exponential functions. (B) Defining each exponential term in model $f_1(t)$ as a monophasic artifact, with the second exponential function triggered by a step function $u(t - t_1)$ (in blue) as the activation function. (C) Refining model $f_1(t)$ to $f_2(t)$ by requiring the decay factors of both exponential b_1, b_2 to share the same value b (in blue) (D) Replacing the step function $u(t-t_1)$ in $f_2(t)$ with a sigmoid function $\sigma(t-t_1)$ (in blue) to satisfy the smooth continuity in BEAM $f_3(t)$.

2) Modified Model Considering Boundary Conditions: Three boundary conditions (the activation function, identical decay rate, and smooth continuity) are introduced in Equation 1 to obtain BEAM, as shown in Figure 2.

a) Activation function: (from Figure 2A to B) Two individual pulses in biphasic stimulation are delivered sequentially, resulting in different starting moment of two artifact components. Here we introduce an activation function $u(t)$ to control the delay of the second artifact element. Then Equation 2 is modified as

$$f_1(t) = f_{\text{Mono}_1}(t) + f_{\text{Mono}_2}(t - t_1) \cdot u(t - t_1), \tag{3}$$

where $t_1 \in \mathbb{R}^+$ is a constant representing the time between two monophasic pulses. The step function $u(\cdot)$ is defined as $u(t - t_1) := (|t - t_1|/(t - t_1) + 1)/2$. The morphology of Equation 3 is shown in Figure 2B.

b) Identical decay rate: (from Figure 2B to C) Both decays occur within the same equivalent electrical circuit of tissue, modeled as an RC network (Appendix-A), in which the passive component exhibits an exponential decay with a time constant equal to the inverse of the product of resistance and capacitance. As a result, parameters b_1 and b_2, share the same decay rate b, i.e.

$$b_1 = b_2 = b. \tag{4}$$

The model morphology refined by this constraint is shown in Figure 2C.

c) Smooth continuity: (from Figure 2C to D) Additionally, the voltage change in human body should be continuous and smooth. It implies the following condition must be satisfied at an arbitrary moment

$$\begin{cases} f(t^+) = f(t^-) \\ f'(t^+) = f'(t^-) \end{cases} \tag{5}$$

where $f'(\cdot)$ denotes the derivative of $f(\cdot)$ and superscripts $-, +$ indicate the left-hand and right-hand limits, respectively. Specifically, the starting time of the artifact is induced by the second pulse, i.e. $t = t_1$. While the exponential term in Equation 3 satisfies the smooth and continuous assumption, the step function $u(t - t_1)$ exhibits a discontinuity at moment t_1.

As proven in Appendix-B, the introduction of step function violates the smooth continuity assumption. Therefore, a sigmoid function $\sigma(t)$ is provided to replace the step function $u(t)$ in Equation 3. It is defined as $\sigma(t) := 1/(1 + \exp(-(t - t_1)/k))$, where k is a factor that controls the time scale. Then BEAM is derived from Equation 3, 4, 5 as

$$f_3(t|a_1, a_2, b, c, t_1, k) = a_1 \cdot e^{-\frac{t}{b}} + a_2 \cdot e^{-\frac{t}{b}} \cdot \frac{1}{1 + e^{-\frac{t-t_1}{k}}} + c. \tag{6}$$

Parameters a_1, a_2, b, c, t_1, k control the position, scale and shape of BEAM. The typical shape of BEAM $f_3(t)$ is shown in Figure 2D.

Note that all boundary conditions are grounded in the settings of electrical stimulation and artifact. Consequently, although developed for biphasic SCS, BEAM does not rely on assumptions about the neural response type may not be limited to SCS.

B. Fitting Method

BEAM $f_3(t)$ is utilized to model the artifact (Figure 1I) from measured signal $V_i(t)$ (Figure 1H), where i is the number of repeated measurements. As illustrated in Figure 1H to J, ideal ECAP signal $f_{\text{ECAP}}(t)$ (Figure 1J) is obtained from denoised waveform $V_i(t)$ by $f_{\text{ECAP}}(t) = V_i(t) - f_3(t)$.

Assuming that ECAP signal $f_{\text{ECAP}}(t)$ is much less than artifact $f_3(t)$ and its fitting error $E(t)$, estimated BEAM $\hat{f}_3(t|\hat{a}_1, \hat{a}_2, \hat{b}, \hat{c}, \hat{t}_1, \hat{k})$ can be deduced from least square method as

$$\arg \min_{\hat{a}_1, \hat{a}_2, \hat{b}, \hat{c}, \hat{t}_1, \hat{k}} E(t) \approx E(t) + f_{\text{ECAP}}(t) = \sum_t (V_i(t) - \hat{f}_3(t))^2$$

where estimated parameters $\hat{a}_1, \hat{a}_2, \hat{b}, \hat{c}, \hat{t}_1, \hat{k}$ are obtained by minimizing the error function $E(t)$. Then measured ECAP signal is estimated as

$$\hat{f}_{\text{ECAP}}(t) = V_i(t) - \hat{f}_3(t). \tag{7}$$

C. Evaluation Method

To assess the performance of ECAP extraction method (Equation 7), we adopt a growth-curve-based approach to quantify both the denoising rate and fidelity rate as evaluating markers. Additionally, we introduce E-score as a comprehensive metric to evaluate extraction efficacy.

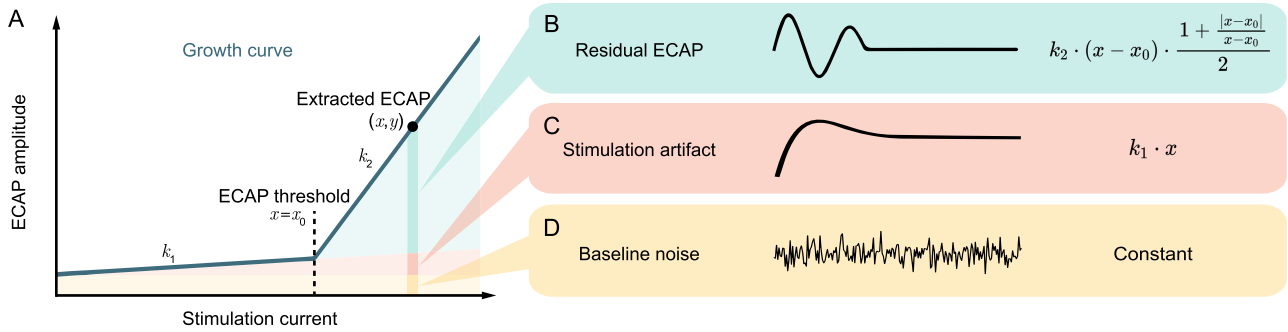


Fig. 3. The growth curve model of extracted evoked compound action potential (ECAP). (A) The growth curve derived from the amplitudes of extracted ECAP under varying stimulation currents. Arbitrary point *(x,y)* on the growth curve represents the magnitude *y* of the extracted ECAP at a fixed stimulation current *x*, comprising (B) residual ECAP after processing, (C) stimulation artifact that has not been removed and (D) the baseline noise including random noise and baseline offset. Constants k_1, k_2 in curve model (A) are different slopes before and after the ECAP threshold x_0,respectively.

1) Growth Curve Model: The growth curve is the estimation of neural activation levels, where the intensity of stimulation current and the amplitude of extracted ECAP are selected as the independent and dependent variables, respectively [22]. Some works refer to it as the "recruitment curve" [23] or "input-output (I/O) function" [24]. This work introduces a growth curve model with two linear segments [8], [24]. The amplitude of extracted ECAP $f_{\text{Ext}}(x)$ in growth curve model consists of residual ECAP $f_{\text{Res}}(x)$ and noise $N(x)$, as illustrated in Figure 3A, that is

$$f_{\text{Ext}}(x) = f_{\text{Res}}(x) + N(x), \tag{8}$$

where x denotes the intensity of stimulation current. Of note, noise $N(x)$ is a non-physiological component that includes stimulation artifact $f_{\text{Arti}}(x)$ and baseline noise $f_{\text{Bas}}(x)$. The residual ECAP does not occur until the stimulation current exceeds threshold. Its activation and linear growth can be described as $f_{\text{Res}}(x) = k_{\text{ECAP}} \cdot (x - x_{\text{Thr}}) \cdot (|x - x_{\text{Thr}}|/(x - x_{\text{Thr}}) + 1)/2$, where x_{Thr} is the current threshold that starts neural response and k_{ECAP} describes the growth rate of residual ECAP (see Figure 3B). Similarly, stimulation artifact exhibits thresholdless linear growth by $f_{\text{Arti}}(x) = k_{\text{Arti}} \cdot x$, where k_{Arti} describes the rate of stimulation artifact growth with current (see Figure 3C). Baseline noise remains constant n as $f_{\text{Bas}}(x) = n$ (see Figure 3D). Then growth curve model in Equation 8 can be written as

$$f_{\text{Ext}}(x) = k_{\text{ECAP}} \cdot (x - x_{\text{Thr}}) \cdot \frac{1 + \frac{|x-x_{\text{Thr}}|}{x-x_{\text{Thr}}}}{2} + k_{\text{Arti}} \cdot x + n. \tag{9}$$

2) Evaluation Metric: To evaluate efficiency of ECAP extraction, we separate neural response and non-physiologic component from measured data via fitting growth curve model. Denoising rate D and fidelity rate F are introduced to compare different denoising methods

$$\begin{cases} D(x) = 1 - \dfrac{N^*(x)}{N(x)} \\ F(x) = \dfrac{f^*_{\text{Res}}(x)}{f_{\text{Res}}(x)} \end{cases} \tag{10}$$

where variables with and without asterisk superscript represent post-denoising and pre-denoising growth curve data, respectively. To jointly quantify denoising and fidelity, we define a single metric E-score $f_{\text{E-score}}$ (the Extracted Score), as the harmonic mean of the denoising rate D and the fidelity rate F

$$f_{\text{E-score}}(x) = 2 \cdot \frac{D(x) \cdot F(x)}{D(x) + F(x)}. \tag{11}$$

This measure evaluates extraction effectiveness while penalizing methods that perform well on one criterion but poorly on the other.

III. Experiment

In this study, we validated BEAM clinically following experiment pipeline (see Figure 4). The performance evaluation of BEAM is compared with other methods.

A. Participants

A 32-year-old female (patient 1) and a 29-year-old male (patient 2) with spinal cord injury were recruited in clinical study. It was registered with ClinicalTrials.gov (NCT04969042) and approved by Beijing Tsinghua Changgung Hospital Ethics Committee (No. 21161-2-02) [25]. Each patient was implanted with a sensing-enabled neurostimulator (Beijing PINS Medical Co., Ltd., model G122RS) and a 32-contact paddle electrode (Beijing PINS Medical Co., Ltd., model L3255). Electrode arrays are placed along the dorsal column of the spinal cord, as shown in Figure 4D-F, the 4×8 contacts span from T12 to L2 spinal segments.

B. Experimental Protocol and Data Acquisition

The experiment started one month after implantation surgery. It comprised 8 data collection sessions during 9 months. To acquire ECAP, participants laid on their back for 3-hour in each data collection session. No unanticipated adverse events were recorded.

In the experiment, command and data communication between host computer and implantable pulse generator (IPG) was transmitted via Bluetooth, as illustrated in Figure 4B. The command defined the electrodes configuration, output waveform and recording setting. The raw data collected neural activity and stimulation artifact.

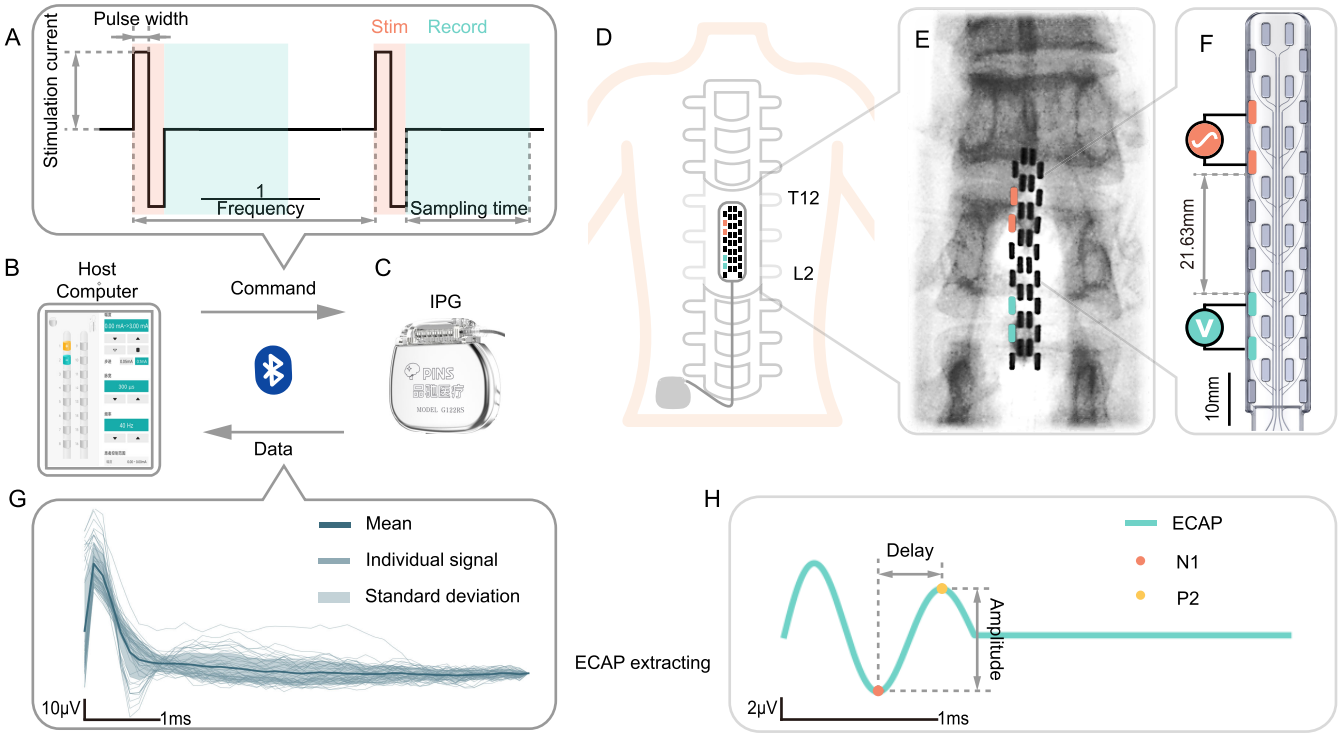


Fig. 4. The pipeline of clinical experiment. (A) Stimulation parameter and recording setting. Stimulation current represents the amplitude of the stimulation pulse, pulse width refers to the duration of monophase pulse and frequency controls the interval between pulse groups. The recording (in light blue) begins immediately after the stimulation (in orange). The record duration is controlled by the sampling time. (B and C) Host computer (B) and implanted pulse generator (IPG) (C) communicates via Bluetooth. The host computer transmits the command of stimulation parameters to the IPG while the IPG sends measured signals back. (D) Stimulation electrodes are positioned in the epidural space of the spinal cord at T12-L2 segment, connecting with the IPG implanted in the lower back. (E) The frontal view of Post-implantation CT image of patient 1 showing electrode placement (in black). Stimulation contacts (in orange) and recording contacts (in light blue) are selected on the rightmost column of contacts. (F) The illustration of electrodes design. (G) Fifty signals were randomly selected from 2084 measured signals under 1 mA, 40 Hz, 300 μs stimulations of patient 1, showing the mean (dark blue line) and the standard deviation (blue shaded) of individual signals (blue line). (H) The illustration of feature definitions of evoked compound action potential (ECAP) after extraction on host computer. The first trough in the extracted ECAP is defined as N1 (orange point), the second peak as P2 (yellow point). The time difference between N1 and P2 is called the ECAP Delay and the peak-to-peak value between N1 and P2 is referred to as ECAP Amplitude.

TABLE I
THE STIMULATION PARAMETERS OF PATIENT 1

Freq (Hz), Sample duration (ms) \ Current[a] (mA)	Pulse width (μs) 100	200	300	400	500
20, 43.75	3.2	1.1	0.85	1.3	0.8
40, 18.75	1.6	1.15	0.85	0.8	0.75
60, 12.5	2.0	1.0	0.95	0.6	0.6
80, 6.25	1.3	0.8	0.75	0.8	0.8
100, 6.25	1.6	1.2	0.75	0.75	0.75

[a] The stimulation current provided in the table represents the maximum value applied during the experiment of patient 1 at the given frequency and pulse width.

Stimulation outputted biphasic pulse (Figure 4A) via adjacent contacts positioned upstream (Figure 4F). Biphasic waveform varied stimulation current (the amplitude of pulse), pulse width (the duration of a monophasic pulse) and frequency (controlling the interval between sequences). The stimulation parameters of patient 1 were shown in Table I. We increased stimulation current every 5 seconds until the participant reported discomfort.

Data were recorded differentially from two adjacent contacts positioned 21.63 mm from the stimulation electrodes, as illustrated in Figure 4F. Data recording was performed in synchronous mode (Figure 4A), i.e., the measurement starts immediately after the end of the stimulation. Recording time window was determined as the maximal integer multiple of 6.25 ms until next pulse. Data sampling rate was 8 kHz. The recording hardware employed a band-pass filter ranging from 0.6 Hz to 4 kHz.

C. Data Analysis

As experimental protocol, we collected a total of 322200 signals under 1140 stimulation parameters from two patients. After excluding abnormal data under unsteady states or suffering from packet loss, the dataset retained 208336 signals under 900 stimulation parameters. The example of exclusion data were shown in Appendix-E.

To reduce the impact of random noise, the measured signals acquired under identical stimulation parameters within the same subject and the same session were averaged to produce a single representative signal. To normalize the data baseline,

each averaged signal segment was shifted by the value of its last data point [26].

The ECAP results were subsequently estimated from each averaged signal using BEAM. The fitting algorithm (Equation 7) was implemented using the optimize.curve_fit function from the SciPy package (version 1.10.1) in Python (version 3.11.3). To compare with BEAM, other methods (single exponential artifact models [14], high-pass filtering [10], ECAP template matching [15]) were selected and implemented.

To evaluate the extraction methods, the growth curve was constructed using Equation 9. As illustrated in Figure 4H, ECAP amplitude was characterized by N1-P2 feature, that was the peak-to-peak amplitude between the first trough (N1) and the second peak (P2). N1 and P2 were extracted from the denoised signals. The peak positions of N1 and P2 were determined from the denoised signals obtained under the highest stimulation current at the same pulse width and frequency, since clear N1 and P2 peaks were often unobservable at lower current or in measured signals. The growth curve was computed by fitting N1-P2 feature under stimulation current. Then, contributions of the neural activation and noise were separated from growth curve as shown in Figure 3B-D. Subsequently, the evaluation metrics were computed using Equations 10 and 11.

Additionally, ablation experiment was conducted to evaluate the necessity of each boundary condition. We compared the extraction result of BEAM with modified model lacking one boundary condition, such as activation function, identical decay rate and smooth continuity, respectively.

Furthermore, to validate BEAM's robustness across neural targets, we applied it to local field potential (LFP) recordings from a previously published deep brain stimulation (DBS) study [27] for stimulation-artifact removal. LFPs were recorded during stimulation at 2 mA, with a 70 μs pulse width and a frequency of 130 Hz.

In clinical data, it is difficult to determine the true value of the ECAP component. Therefore, synthetic data were used to validate the BEAM denoising method and the Growth Curve evaluation approach. The synthetic data were generated using stimulation artifacts collected in physiological saline under 0.4–1.2 mA, 40 Hz, 300 μs stimulation, with an ECAP component simulated by a 1.5-cycle sinusoidal signal superimposed on the artifact. The ECAP component was set to begin appearing at a stimulation intensity of 0.7 mA, with the N1-P2 peak-to-peak value linearly increasing to 0.12 times the peak-to-peak range of the artifact at 1.2 mA.

IV. Results

Figure 4G presents 50 raw signals acquired during identical stimulation parameter (40Hz, 300us, 1mA), as well as its average and variance. Those signals exhibit consistent shape, with 3 stages (initial increase, subsequent decrease and final stabilization). Comparing the mean signal and the raw data, the fluctuations of individual signals are eliminated by averaging operation.

Figure 5 shows the validation of BEAM on synthetic data, achieving an E-score of 0.87 with a denoising rate of 83% and a signal fidelity rate of 92%. The quantitative metrics derived from the growth curve are also highly consistent with those computed using the ground truth, with differences within 5%.

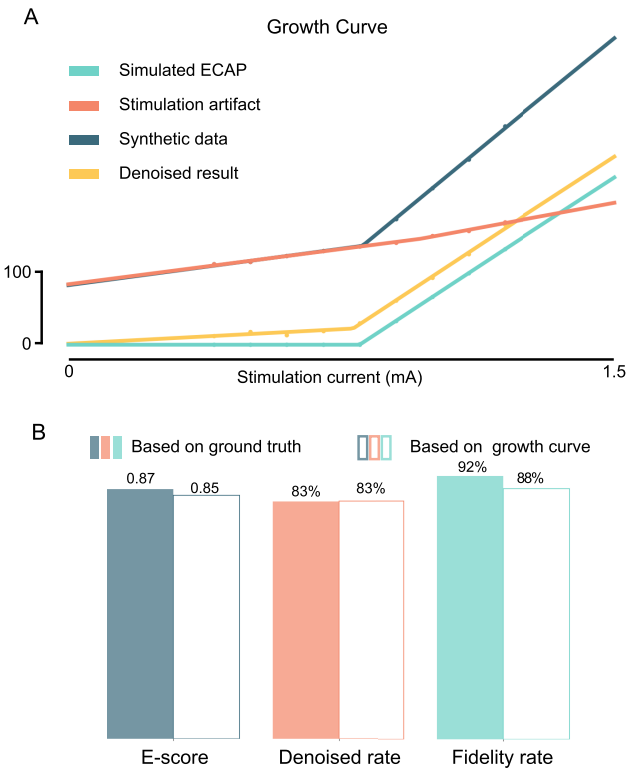


Fig. 5. BEAM evaluated on synthetic data. (A) Growth curves for the simulated ECAP, the stimulation artifact recorded in saline, and the composite synthetic signal, the BEAM-denoised output. (B) E-score (dark blue), denoised rate (orange), fidelity rate (light blue) computed based on either the ground truth or the growth curve.

Figure 6 shows the ECAP extracted using BEAM method from patient 1. Figure 6A to F compare measured results before and after BEAM processing varying amplitude (Figure 6A and D), frequency (Figure 6B and E), pulse width (Figure 6C and F). The measured signals (Figure 6A to C) show a maximum amplitude on the order of 100 μV, gradually diminishing to zero within 3 ms. These signals display a single peak within the first millisecond. The extracted ECAP signals (Figure 6D to F) show a maximum amplitude on the order of 10 μV, reaching a steady state within 2 ms. Morphologically, most extracted ECAP signals exhibit a positive-negative-positive triphase sequence. The triphase sequences occur within the first 2 ms. The first trough (N1) appears between 0.375 and 0.5 ms. Based on this latency, the calculated conduction velocity ranges from 43.26 to 57.68 m/s, which falls within the typical range of Aβ fibers (35–75 m/s). Figure 6G to I presents the quantitative metrics of BEAM's denoising performance across different stimulation parameters. In most cases, BEAM maintains a high E-score. Even under the least favorable conditions, it still achieves an E-score of 0.55, with a denoising rate of 39% and a signal fidelity rate of 88%. Figure 6J to L shows the magnitudes of ECAP and the value of two exponential terms a_1 and a_2 in fitting model

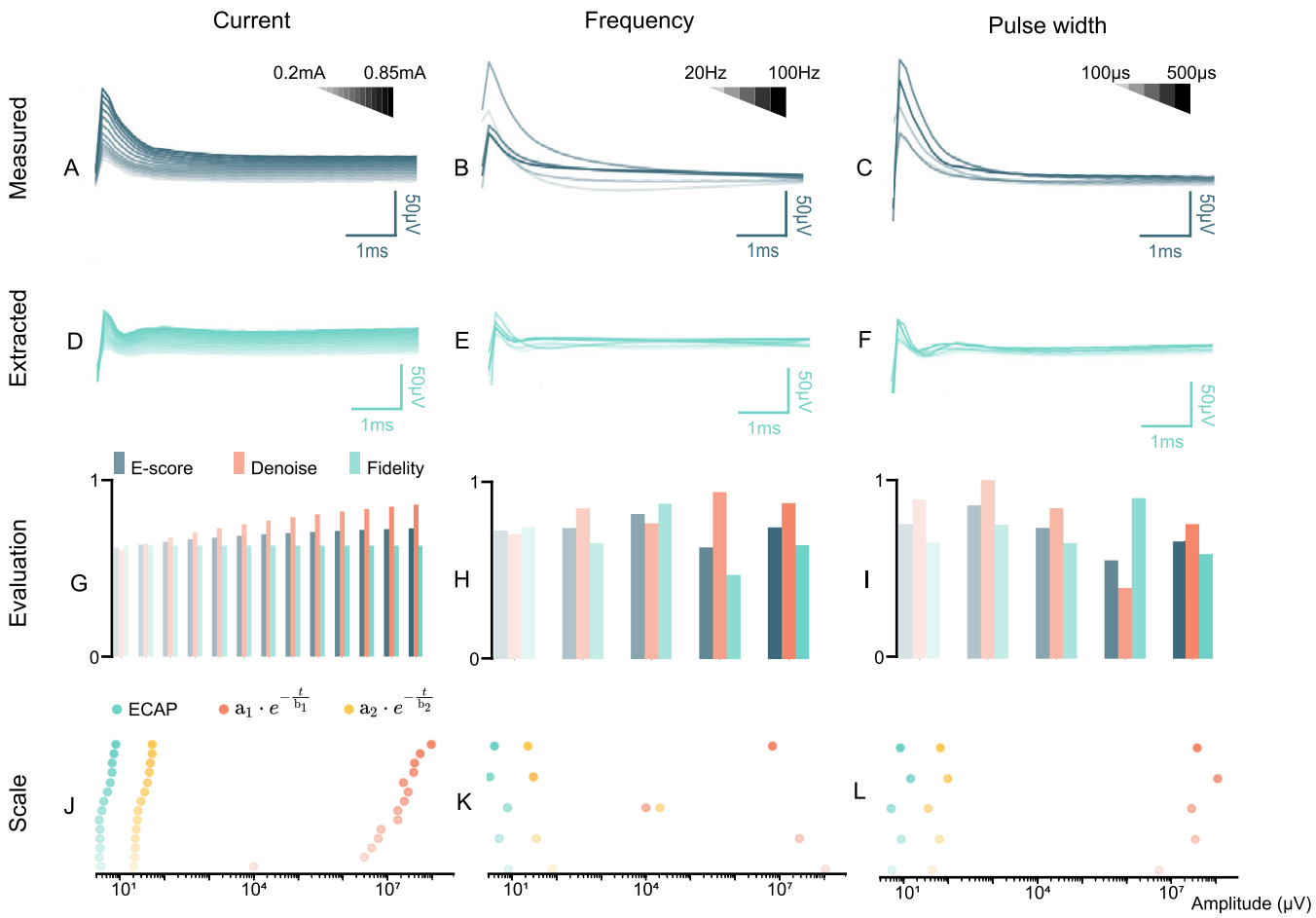


Fig. 6. Extracted results of the biphasic electrostimulation artifact model (BEAM) under different stimulation parameters from patient 1. Measured signal (A, B and C) and extracted evoked compound action potential (ECAP) (D, E and F) are plotted. (G, H and I) E-score (deep blue), denoised rate (orange), fidelity rate (light blue). (J, K and L) The scale of ECAP features (in light blue) processed by BEAM and the amplitude of each exponential term (in orange and yellow) in BEAM after fitting. (A) and (D) show responses with stimulation current stepped from 0.2 to 0.85 mA in 0.05 mA increments. To enhance the separation between different signals, a bias was applied to each signal only in the plot,, increasing by 2 μV with each parameter increment. (B) and (E) display results with stimulation frequency stepped from 20 to 100 Hz in 20 Hz increments. (C) and (F) illustrate responses with pulse width stepped from 100 to 500 μs in 100 μs increments. The progression of parameter values is represented by color intensity from light to dark. (J, K and L) The height and color intensity correspond to the varying parameter values. The x-axis shows the magnitude on a logarithmic scale.

(Equation 7). Notably, the magnitudes of both exponential term are at least an order of magnitude larger than the ECAP signal. In most case, the magnitude of the first exponential term exceeds that of the second term by more than six orders of magnitude. Furthermore, as shown in Figure 6J, both a_1 and a_2 increase with rising stimulation current.

The results of ablation experiments are exhibited in Figure 7. Figure 7A shows the ECAP waveforms extracted using BEAM (A_1) and modified BEAM eliminated boundary of smooth continuity condition (A_2), identical decay rate (A_3), activation function (A_4), respectively. The results extracted by classical model are also presented (A_5). BEAM (A_1) and the model without smooth continuity condition (A_2) are able to recover the triphasic waveform at high stimulation currents. Model without identical decay rate (A_3) or activation function (A_4) extracts biphasic waveforms resembling those of the classical double exponential model (A_5). The amplitude of each peak decreases sequentially over time for all models, except for the model without the smooth continuity condition (A_2). In this case, the extracted results exhibit larger peak amplitudes for later-occurring peaks, which is likely an overestimate causing by discontinuities. Figure 7B to D presents the quantitative metrics for the extraction results using models with various boundary conditions. The BEAM method achieves the highest E-score of 0.72. Removing any boundary condition results in a decrease in the E-score. Of note, while the fidelity rates of other models increase, they all exceed a value of 1.

Figure 8 compares the extracting performance of BEAM with other algorithms described in Section III-C, including single exponential artifact models, high-pass filtering and ECAP template matching. Figure 8A compares the extracted results from different methods, where a distinct neural signal component is not visible until the current strength of 1.0 mA. The extracted results from BEAM (Figure 8A_1) exhibit a distinct triphasic morphology at larger currents, where the

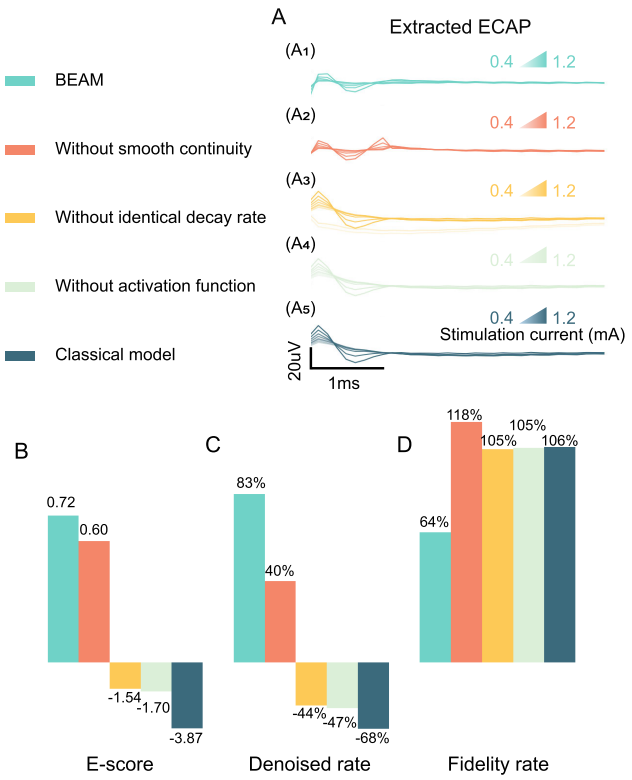


Fig. 7. Extracted evoked compound action potential (ECAP) from the ablation experiment. Each boundary condition was individually removed from the biphasic electrostimulation artifact model (BEAM). (A) The extracted ECAP waveforms under different models, including BEAM (A_1), BEAM without smooth and continuity constraint (A_2), BEAM without identical decay rate constraint (A_3), BEAM without activation function constraint (A_4) and classical double exponential model (A_5). The stimulation currents are indicated by transparency levels. (B, C and D) Comparison of quantitative metrics. The E-score (B) represents the harmonic mean of the denoising rate (C) and fidelity rate (D).

amplitude of each peak in the triphasic waveform increases with rising stimulation intensity. Conversely, the results of single exponential artifact model method (Figure 8A_2) exhibit an exponentially decaying shape at smaller current and a triphasic component on the decaying curve at larger currents. The high-pass filter method (Figure 8A_3) extracts curve that peaks and troughs are slightly shifted to the left. The template matching method (Figure 8A_4) loses the second and third peak in baseline fluctuations. Figure 8B evaluates results of different extracting methods via growth curve. All growth curves exhibit inflection points around 0.9 mA for patient 1 and 0.7 mA for patient 2, representing the ECAP threshold. The subthreshold growth curves derive from BEAM and template matching methods remain nearly horizontal, with the latter showing a near-zero bias. The suprathreshold growth curves from all methods exhibit noticeable increase in slope. The template matching method exhibits the lowest slope among all methods. Figure 8C to E summarize the quantitative metrics of the extraction results based on the growth curve model. The BEAM method achieves the highest performance with 0.72 E-score in patient 1 and 0.92 E-score in patient 2. Other methods may excel in noise denoising or ECAP fidelity, but fail in another term. For example, the template matching method removes 99% noise in patient 1 while reserves only 10% neural component.

Figure 9 assesses the robustness of BEAM method for denoising stimulation artifacts in long-term ECAP data from SCS and local field potential (LFP) data from DBS [27]. The extraction performances across nine-month application are shown in Figure 9A to C. BEAM maintains E-scores (Figure 9A) in the range of 0.53-0.81 without significant degradation. Figure 9D and E display the extraction result of BEAM in DBS. The measured signals (dark blue) from multiple pulses are concatenated and displayed in the time domain (Figure 9D), revealing a series of periodic, millivolt level negative peaks. These peaks are significantly suppressed in the extracted result (light blue). In the frequency domain (Figure 9E), the measured signal exhibits a pronounced peak at 130 Hz, corresponding to the stimulation frequency. This peak is attenuated after extraction, while low-frequency components from 0 to 40 Hz are largely preserved.

V. Discussion

In this study, we introduce BEAM for extracting ECAP in neuromodulation. We validate its effectiveness clinically in both SCS and DBS scenarios. Additionally, BEAM is evaluated by E-score and growth curve, which performs comprehensive advantages comparing with other methods.

A. Evaluation of ECAP Extraction

Although several ECAP extraction methods have been proposed, few studies evaluated the performance of different ECAP extraction approaches [10], [28]. Because clinical ECAP lacks a gold standard, it is hard to quantify extraction result. Previous work estimates ECAP amplitude from raw data via growth curve to calculate denoise rate and fidelity rate, making it possible to compare ECAP extraction result from different methods [10]. However, neither denoising rate nor fidelity rate is able to comprehensively evaluate extraction performance, as shown in Figure 8C to E. Extraction method may achieves well in one index but fails in another index. To address this limitation, we introduce the E-score metric that balances the denoising rate and fidelity rate through their harmonic mean, providing a single comprehensive metric (as demonstrated in the Appendix-C).

Additionally, the growth curve method provides a way to verify the presence of neural components. Previous studies [8], [29] have shown conduction delays across two recording contacts indicating a “traveling” action potential. However, this requires sufficient propagation space and cannot be applied in a single-electrode setup. The shape of the growth curve helps distinguish between neural and non-neural components, based on the neural threshold. The neural activity appears only after reaching a certain stimulation threshold, while the non-neural component is always present.

It should be noted that using the growth curve to quantify ECAP extraction performance requires data collected across multiple stimulation intensities. the metric cannot be derived

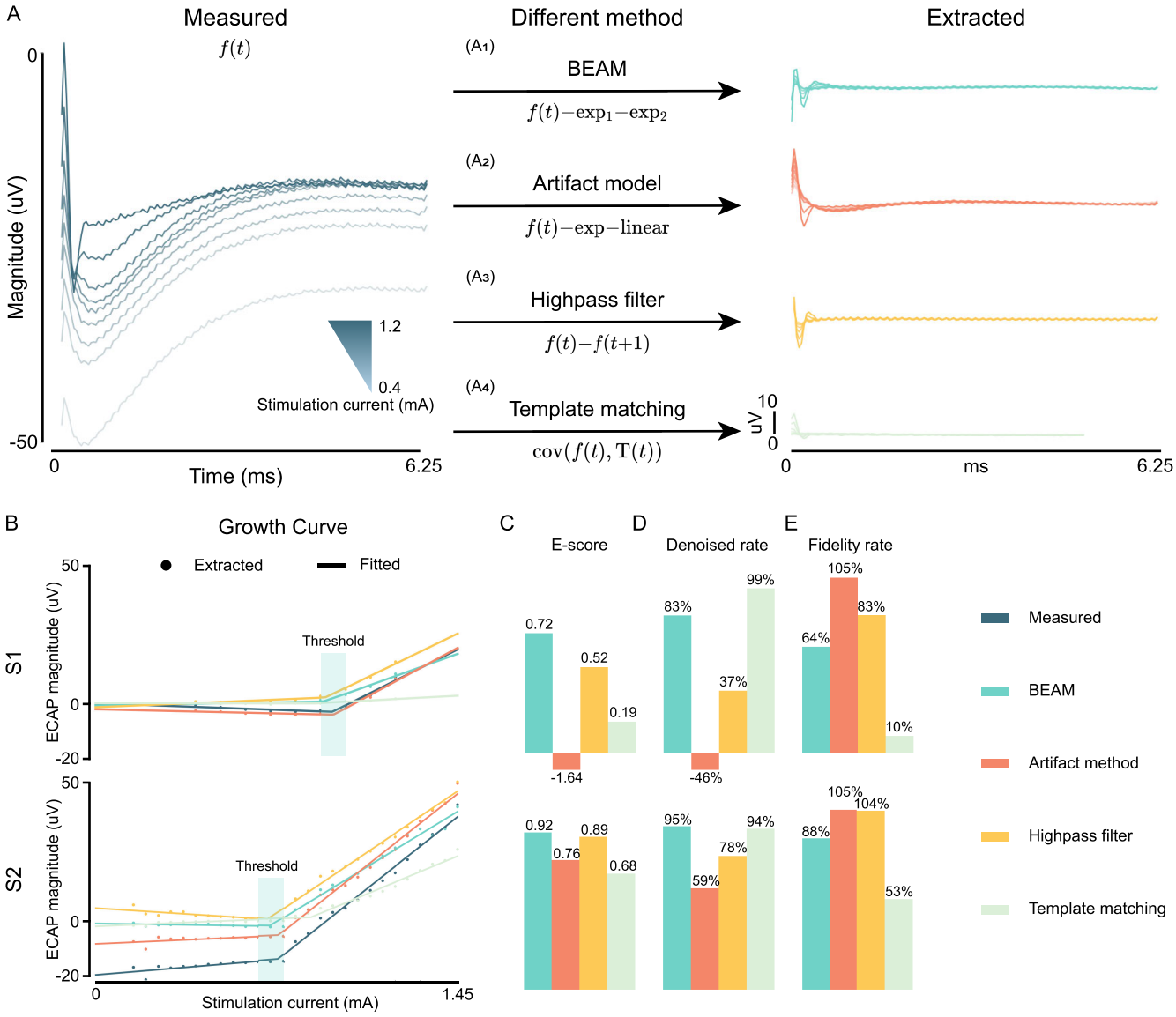


Fig. 8. Comparison of different artifact removal algorithms on measured signals from patient 1 (S1) and patient 2 (S2). (A_1) Biphasic electrostimulation artifact model method (BEAM, in light blue) fits artifact using two exponential terms with constraints. (A_2) Artifact model (in orange) fits artifact using an exponential term combined with a linear term. (A_3) Highpass filter (in yellow) removes low-frequency components by applying differentiation. (A_4) Template matching (in light green) extracts the evoked compound action potential (ECAP) by correlating the measured signal with a predefined template. (A) The measured signals (in dark blue) and the corresponding extracted ECAP signals from S1 are shown at varying stimulation currents (ranging from 0.4 to 1.2 mA in 0.1 mA increments) with transparency levels indicating the intensity of each stimulation. (B) The extracted ECAP results (dots) and fitted growth curves (lines) of S1 and S2. The ECAP threshold is observed at a stimulation current around 0.9 mA for S1 and 0.7 mA for S2. (C, D and E) Comparison of quantitative metrics from S1 and S2. The E-score (C) represents the harmonic mean of the denoising rate (D) and fidelity rate (E).

from comparing a single ECAP signal before and after processing, and the data must span the ECAP threshold. In addition, its application demands strict control of experimental variables, such as maintaining a stable posture, so that the stimulation current remains the only factor that varies.

B. Validity and Generality of BEAM

The proposed BEAM method extracts ECAP effectively under diverse stimulation conditions. As shown in Figure 6, BEAM fits stimulation artifacts well and reduces artifact amplitude significantly under various stimulation currents, frequency and pulse width. It aligns with the understanding that stimulation artifacts are substantially larger than neural responses in spinal cord stimulation scenario [8], [30]. The extracted signals exhibit a triphasic morphology within 2 ms, which is consistent with ECAP patterns reported in previous studies [8], [9]. Furthermore, the results reveal stable neural responses above the stimulation threshold that increasing with stimulation current, matching patterns in the literature [8], [16]. Quantitatively, BEAM method achieves the highest E-score compared to other methods, reflecting its optimal

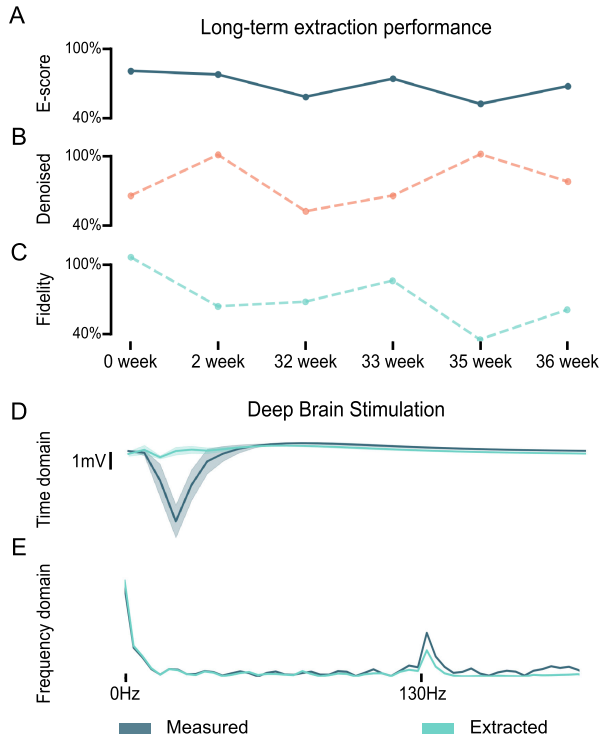


Fig. 9. The robustness of Biphasic electrostimulation artifact model (BEAM). (A, B and C) The stability in long-term clinical condition. Extraction performances across nine-month application are evaluated by E-score (A), denoised rate (B) and fidelity rate (C). (D and E) Generality in deep brain stimulation. The mean (line) and standard deviation (shaded area) of measured signal (dark blue) and extracted result (light blue) are shown in the time domain (D) and frequency domain (E).

balance between noise reduction and preservation of neural components.

The BEAM method extracts ECAP stably across patients, time and scenarios. Firstly, BEAM method performs the highest E-score in both patients(n= 2). Also, it maintains robust denoising performance over a nine-month period, whose duration exceeds most clinical ECAP studies [8], [10], [32]. Previous study typically rely on data from semi-implanted electrodes collected near the end of commercial trials or before trial lead removal, which limited their evaluation of extraction methods for long-term use.

In addition to the SCS scenario, BEAM also removes stimulation artifacts effectively in DBS while preserving the critical frequency information necessary for analyzing LFP [19], [33]. Spectral information is crucial for understanding brain network dynamics, particularly within the 5–35 Hz range, encompassing the theta, alpha and beta bands [34]. As shown in Figure 9D and E, BEAM significantly attenuates the 130 Hz stimulation frequency while preserving the 5–35 Hz band. It may result from origination of BEAM without site-specific assumptions, enhancing its versatility for removing stimulation artifacts across diverse therapies. In contrast, previous studies focus on specific applications, with limited validation on clinical data from diverse scenarios [10], [15], [35].

In addition to the conditions experimentally verified above, other factors may also influence the ECAP signal, such as contact selection, body posture, and stimulus pulse type. Although corresponding experiments were not performed in this work, we provide a theoretical analysis of BEAM's potential applicability under such conditions. The BEAM algorithm can still accurately model and subtract the stimulation artifact if two assumptions are satisfied: (1) the artifact morphology remains unchanged, or can be sufficiently described through parameter adjustments within BEAM; and (2) the artifact amplitude is substantially larger than the ECAP component, allowing the fitting method to remain effective. Changes in ECAP morphology do not affect BEAM's denoising performance, as BEAM models the artifact without making any assumptions about the neural component.

Regarding the influence of referencing strategy and the distance between stimulation and recording electrodes on the ECAP signal, Verma et al. [31] conducted a detailed investigation and found that while the ECAP waveform shape varies considerably across configurations, the artifact waveform remains consistent. Based on these findings, we infer that BEAM has the potential to adapt to different electrode configurations, as long as the artifact amplitude remains substantially larger than the ECAP component.

Similarly, for ECAP recordings obtained under different body postural, the relative positions of the recording and stimulation contacts remain fixed, resulting in a consistent artifact morphology. Therefore, BEAM is expected to yield comparable performance under these conditions.

Also, BEAM was originally designed to handle artifacts from biphasic stimulation, while it has the potential to generalize to other stimulation paradigms, such as monophasic or asymmetric stimulation. This can be achieved by adjusting the parameters a_1, a_2, t_1 within BEAM in Equation 6, allowing artifact fitting for any stimulus composed of no more than two square-wave phases.

Although BEAM can adapt to a wide range of settings, it relies on the assumption that the amplitude of the artifact component is significantly greater than that of the ECAP component. When the data are acquired from recording contacts located far from the stimulation contacts, such as in longitudinal dual-lead implantations, the performance of BEAM may degrade substantially or even fail. Another consideration is that the changes in ECAP morphology do not affect BEAM's denoising performance, averaging as a preprocessing step may eliminate differences in the actual ECAP component, which could be a limitation when focusing on the ECAP morphology.

The evidence above supports that BEAM is capable of accurately fitting stimulation artifacts without underfitting. Further, we demonstrate that BEAM avoids overfitting by examining fitting parameters and ablation experiment. The amplitude coefficients of two exponential terms in BEAM are significantly larger than the neural response, highlighting their critical role in capturing artifact characteristics accurately. Although the amplitude of the first term is substantially larger than that of the second in Figure 6J, K and L, which is in contrast with the assumption of identical reverse pulses. The difference is likely due to simplifications where the

second term's amplitude accounts for time-shifted factors in Equation 3. Additionally, ablation experiment confirms the necessity of each boundary condition that constrains BEAM model. Removing any boundary condition markedly disrupt the characteristic triphasic waveform and reduce the E-score.

C. Application and Clinical Value of BEAM

ECAP-based closed-loop SCS technology has superior outcomes in chronic pain management compared to traditional methods [6], [36]. However, the universal method still remain controversial to obtain accurate ECAP signals under clinical conditions. Some studies propose separating stimulation artifacts based on differences in propagation speed—light-speed for artifacts versus neural conduction speed for ECAP signals [8], [29]. While effective in theory, these methods require propagation distances that are impractical in therapeutic settings and often necessitate an additional implanted electrode. BEAM provides a novel solution for ECAP extraction in spatially constrained environments. Its ability to operate effectively within such limitations highlights its potential to advance clinical applications in neuromodulation.

Furthermore, electrical stimulation is known to drive neural plasticity, which can lead to changes in ECAP signals over time [37], [38]. However, the adaptability of ECAP extraction algorithms to these changes remains unexplored. BEAM's stability stems from its foundation in characterizing stimulation artifact processes, which remain consistent over long-term implantation. This robust framework highlights BEAM's potential in long-term clinical applications.

D. Limitations

Several limitations exist in this study. First, we tested the ECAP extraction method in only two subjects. Future work will aim to validate generality across diverse datasets. Additionally, the BEAM method requires high computational cost of fitting, which may limit its use in real-time closed-loop clinical applications. Despite these limitations, the extraction methods and evaluation metrics presented here are valuable tools for advancing the understanding and research of neural responses in neuromodulation.

VI. Conclusion

This study proposes a universal BEAM method to extract ECAP in neuromodulation scenarios. Our experiments demonstrate that BEAM has advantages in biphasic stimulation artifact removal for clinical data processing. Its robustness is validated over temporal variations, individual differences and diverse neuromodulation therapies. BEAM enhances the accuracy of ECAP extraction to monitor post-stimulation neural responses, providing insights into closed-loop control and neuromodulation mechanism research.

Appendix

A. The Artifact of Monophasic Stimulation

The electrical environment inside the human body can be modeled as an impedance-capacitance circuit. After the pulse is delivered, the capacitor stores charge. When the stimulus source is disconnected from the circuit, the discharge begins, as shown in Figure 10.

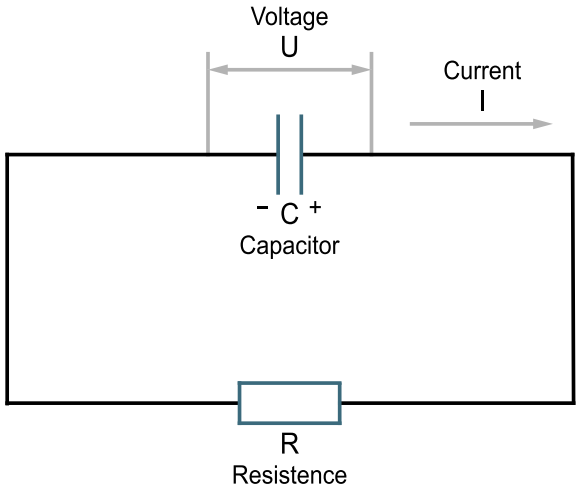


Fig. 10. The discharge circuit model of the human body when stimulus source disconnection. The capacitor C represents charge storage, generating a voltage U, which discharges through the resistance R, resulting in an observable current I.

The artifact originates from the charge carried by the capacitor, which is gradually released in the circuit. The voltage can be expressed as the ratio of the charge to the capacitance $U(t) = Q(t)/C$, where t is time, function $U(t)$ and $Q(t)$ represent the time-dependent voltage and charge, and C represents the capacitance. The current in the circuit is determined by the rate of change of the charge $I(t) = -dQ(t)/dt$, where $I(t)$ represent the current. Additionally, the current can also be expressed as the ratio of the voltage to the resistance $I(t) = U(t)/R$, where R represents the resistance of the circuit. By combining these relationships, the differential equation for the voltage can be derived

$$\frac{dU(t)}{dt} = -\frac{U(t)}{R \cdot C}$$

By solving this differential equation, the voltage as a function of time is described as

$$U(t) = e^{-\frac{t}{RC}} + a$$

where parameter a is a constant. Consequently, the artifact of monophasic stimulation is in the form of decaying exponential.

B. The Activation Function

The activation function defines the onset time of the artifact generated by the second pulse. In the initial model, a step function is utilized; however, this introduces a discontinuity at $t = t_1$. To satisfy the smoothness and continuity, the model parameters must meet additional constraints. Here, these conditions are derived based on the model $f_2(\cdot)$ in Figure 2, which represents the latest version of the model prior to incorporating the continuity and smoothness constraints.

By applying the continuity condition, we have the following equation

$$f_2(t_1^+) = f_2(t_1^-)$$

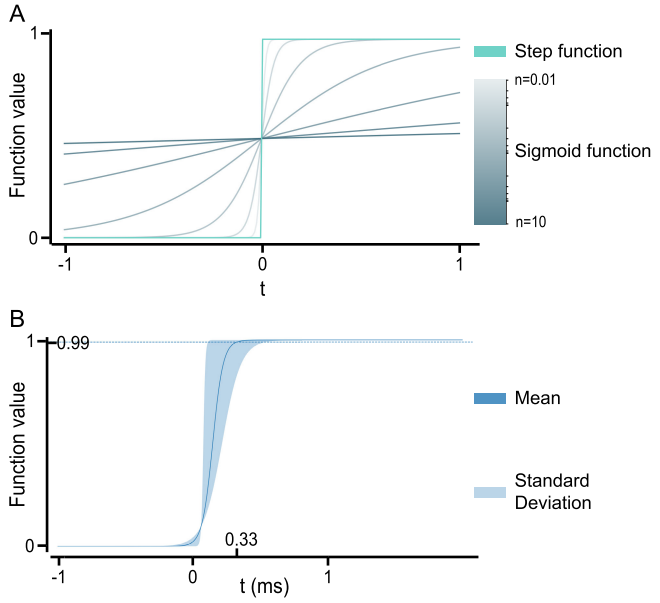


Fig. 11. (A) Visualization of sigmoid functions with varying parameter n, alongside the step function. Sigmoid functions with larger n are depicted with darker lines, while the step function is represented by the light blue line for comparison. (B) The sigmoid component in the fitted results under various stimulation parameters. The mean function value reaches 0.99 at 0.33 ms.

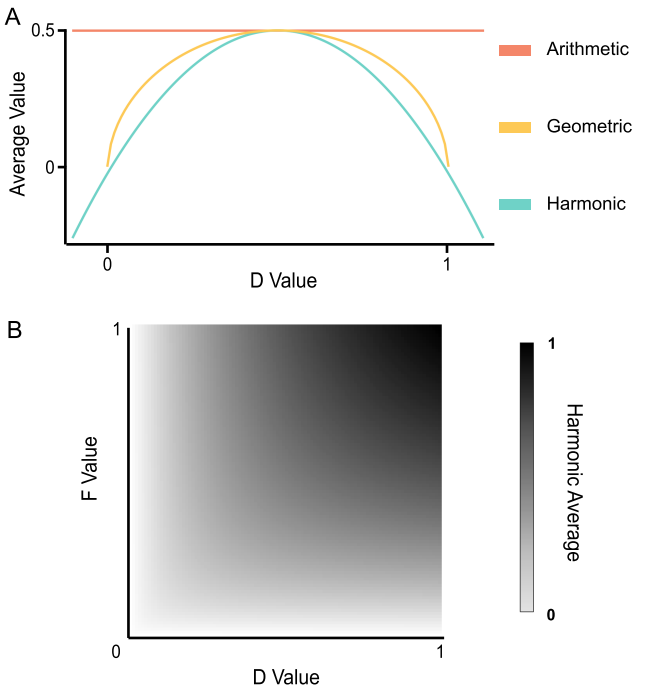


Fig. 12. The average performance cross various component value. (A) The result of different averaging methods under the constraint $D + F = 1$, including arithmetic mean (in orange), geometric mean (in yellow), harmonic mean (in blue). (B) The harmonic mean metric for different combinations of D and F. The average values are represented by color depth from light to dark.

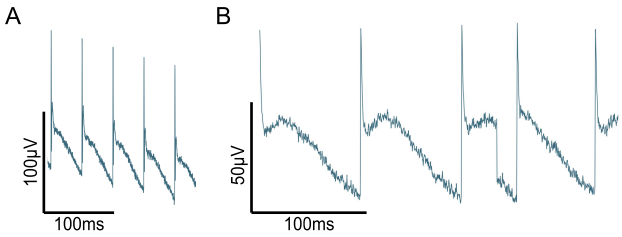


Fig. 13. Example of exclusion data. (A) abnormal data under unsteady states with baseline decreasing. (B) abnormal data suffering from packet loss.

where equation $f_2(\cdot)$ is fully defined in Figure 2, the superscript $-$ indicates the left-hand limit and the superscript $+$ indicates the right-hand limit. From this condition, we obtain

$$c_2 = -a_2 \cdot e^{-b \cdot t_1} \tag{12}$$

where the definitions of a_1, b, c_2 are consistent with those in Equation 2.

Additionally, by enforcing the smoothness condition, we arrive at

$$f_2'(t_1^+) = f_2'(t_1^-)$$

where $f'(t)$ denotes the derivative of $f(t)$. Substituting $f_2(\cdot)$ into this condition and solving it simultaneously with Equation 12, we derive

$$-a_2 \cdot b \cdot e^{b \cdot t_1} = 0$$

Setting either a_2 or b to zero would make the model $f_2(\cdot)$ meaningless. Therefore, the step function is not appropriate to describe the delay of the second exponential component in the artifact.

To satisfy the smoothness and continuity assumption, the sigmoid function is introduced to replace the step function. The general expression for the sigmoid function is as

$$\sigma(t) = \frac{1}{1 + e^{-\frac{t}{n}}} \tag{13}$$

Obviously, the sigmoid function is continuous and smooth everywhere. The shape of the sigmoid function is determined by $n \in \mathbb{R}_+$. Figure 11A shows the shapes of the sigmoid function for different values of n.

When n approaches zero, the shape of the sigmoid function gradually approximates that of the step function. Therefore, using the sigmoid function as an activation function can describe the starting time of the artifact from the second pulse while maintaining the smoothness and continuity of the model.

Figure 11B shows the sigmoid component in the fitted results under various stimulation parameters. The mean function value reaches 0.99 at 0.33 ms, while in the denoised results, the N1 peak appears between 0.375–0.5 ms and the P2 peak around 0.875 ms. This temporal separation excludes the possibility of the sigmoid function causing an overestimation of the ECAP features.

C. The Average Method of Denoise Rate and Fidelity Rate

To balance the denoising rate and fidelity rate, we employ an averaged metric to characterize both aspects. As shown in Figure 8A, a low denoising rate or a low fidelity rate will fail to extract ECAPs effectively, regardless of the success of

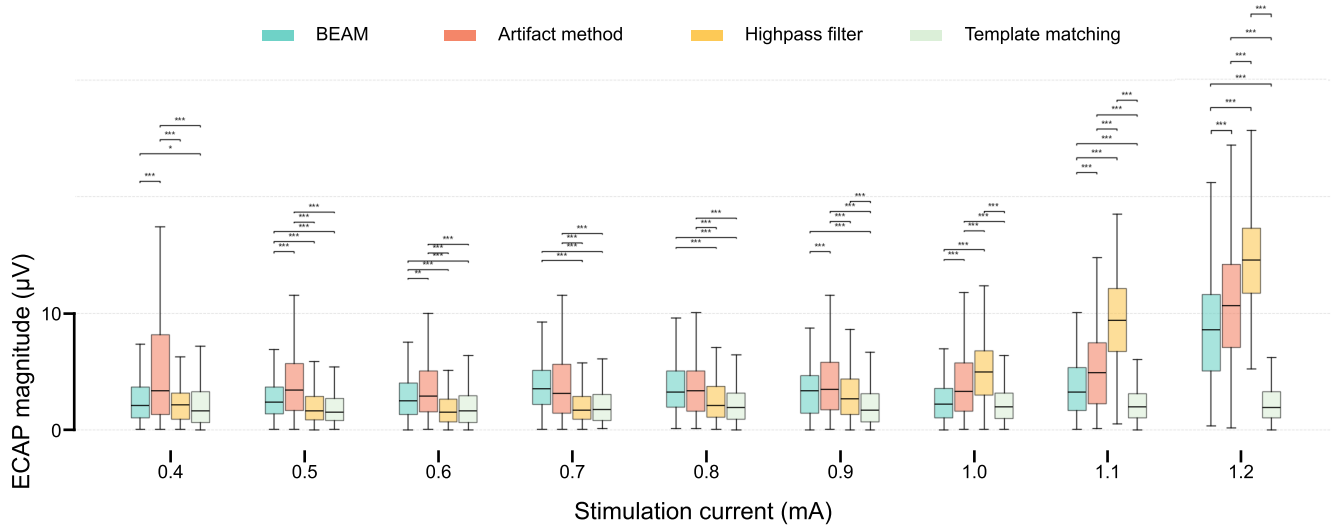


Fig. 14. Statistical test results for different methods. No bar indicates a non-significant difference ($p \geq 0.05$), * denotes $p < 0.05$, ** denotes $p < 0.01$, and *** denotes $p < 0.001$.

the other metric. To address this, we compare three different averaging methods for constructing a balanced metric between the denoising rate D and fidelity rate F, include arithmetic mean $(D + F)/2$, geometric mean $\sqrt{D \cdot F}$, harmonic mean $D \cdot F/(D + F)$. These methods are symmetric with respect to each value being averaged.

Figure 12A illustrates the average results for various combinations of denoising rate and fidelity rate under the constraint $D + F = 1$. The arithmetic mean remains constant at 0.5, failing to reflect the balance between the two metrics. In contrast, geometric mean and harmonic mean achieve their highest scores when $D = F$ and approach zero when either metric is low. When D or F falls outside the range $[0, 1]$, geometric mean yields negative values, indicating errors in ECAP extraction, while harmonic mean becomes undefined in the real number set.

Given these considerations, the geometric mean is selected as the metric to balance denoising rate and fidelity rate. The resulting measure is referred to as the E-score. Figure 12B presents the distribution of E-score across various combinations of denoising rate and fidelity rate.

D. Stimulation Parameter of Patient 2

The stimulation parameters of patient 2 were shown in Table II.

E. Example of Exclusion Data

Figure 13shows excluding abnormal data under unsteady states or suffering from packet loss.

F. Statistical Test for Different Methods

Figure 14shows the statistical results of the ECAP magnitudes extracted by different methods under various stimulation current. The stability of BEAM under the same stimulation parameters is comparable to that of other methods, which may be attributed to the high variability of single-trial signals without averaging. In most cases, BEAM is statistically consistent with the other methods, but it achieves a higher E-score, as demonstrated in Figure 8C.

TABLE II
THE STIMULATION PARAMETERS OF PATIENT 2

Current[a] (mA) / Freq (Hz), Sample duration (ms)	Pulse width (μs)				
	100	200	300	400	500
20, 43.75	4.4	2.3	2	1.6	1.4
40, 18.75	2.5	2.05	1.45	1.4	1.3
60, 12.5	2.9	1.8	1.35	1.35	1.15
80, 6.25	4.5	2.4	1.6	1.65	1.15
100, 6.25	2.5	1.4	0.95	0.8	0.75

[a] The stimulation current provided in the table represents the maximum value applied during the experiment of patient 2 at the given frequency and pulse width.

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